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1 from sklearn.feature_extraction.text import TfidfVectorizer
2 from sklearn.metrics.pairwise import cosine_similarity
3 documents = [
4     "machine learning is useful",
5     "machine learning uses data",
6     "python is useful for machine learning",
7     "data science uses python"
8 ]
9 query = input("Enter search query: ")
10 vectorizer = TfidfVectorizer()
11 tfidf = vectorizer.fit_transform(documents + [query])
12 similarity = cosine_similarity(tfidf[-1], tfidf[:-1]).flatten()
13 ranking = similarity.argsort()[::-1]
14 print("\nDocument Ranking:")
15 for i in ranking:
16     print("Document", i + 1, "Score:", round(similarity[i], 3))
17     print(documents[i])
```

Enter search query: *machine learning*

Document Ranking:

Document 2 Score: 0.573

machine learning uses data

Document 1 Score: 0.573

machine learning is useful

Document 3 Score: 0.421

python is useful for machine learning

Document 4 Score: 0.0

data science uses python

